



# **Master Presenter's Manual**

*for*

**Ethical Hacking**

**Curriculum Code: OV-1525**

**Release Date: June, 2024**  
**Version 1.0**

Amendment Record

| Version No. | Effective Date | Change | Replaced Pages |
|-------------|----------------|--------|----------------|
| 1.0         | June 2024      | New    | 0              |

## Table of Contents

| S# | Details                                          | Page No. |
|----|--------------------------------------------------|----------|
| 1. | <a href="#">Introduction</a>                     | 1        |
| 2. | <a href="#">Information on Module Allocation</a> | 1        |
| 3. | <a href="#">Week-wise Session Schedule</a>       | 2        |
| 4. | <a href="#">Session Coverage</a>                 | 3        |
| 5. | <a href="#">Deliverables</a>                     | 6        |
| 6. | <a href="#">Library References</a>               | 7        |
| 7. | <a href="#">Evaluation Strategy</a>              | 8        |
| 8. | <a href="#">Classroom and Lab Pre-requisites</a> | 9        |

## 1. Introduction

This Presenter's Manual is to be used for the **Ethical Hacking** course.

After the completion of this course, students will be able to:

- ❖ Learn the foundations of ethical hacking
- ❖ Learn how to safeguard digital systems and networks
- ❖ Learn the fundamentals of Cybersecurity principles
- ❖ Understand the essential concepts of penetration testing and vulnerability assessment
- ❖ Learn the methodologies and tools used by ethical hackers to identify and mitigate potential security risks

## 2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on the module along with a break-up of the hours being spent on the module.

This course is based on theory and practice lab session. A session is of two hours duration.

### Module Wise Session Allocation

| Module          | Online Hours | Skill Code |
|-----------------|--------------|------------|
| Ethical Hacking | 40           | 1336       |

## Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

### Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

| Icons                                                                               | Feature - Description/Functionality                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Download Book</b> - Student has the option to download the subject related e-book and read offline.                                                    |
|    | <b>Glossary</b> - Student can access a list of subject related specialized words with their definitions.                                                  |
|    | <b>FAQ</b> - Student can access frequently asked questions and their answers.                                                                             |
|    | <b>Practice 4 Me</b> - Student can test and evaluate their understanding of module related topics.                                                        |
|   | <b>Work Assignments</b> - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.                |
|  | <b>References</b> - Student can access additional subject related material for reading.                                                                   |
|  | <b>Feedback</b> - Student can provide feedback on the course material.                                                                                    |
|  | <b>Ask to Learn</b> – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head. |

The table below shows the details of the module, the number of hours being spent on the module along with a break-up of the hours being spent on each session.

### 3. Week Wise Schedule

| Topic \ Week    |    | 1   | 2   | 3    | 4    | 5    |
|-----------------|----|-----|-----|------|------|------|
| Ethical Hacking | D1 | TL1 | TL5 | TL9  | TL13 | TL17 |
|                 | D2 | TL2 | TL6 | TL10 | TL14 | TL18 |
|                 | D3 | TL3 | TL7 | TL11 | TL15 | TL19 |
|                 | D4 | TL4 | TL8 | TL12 | TL16 | TL20 |

Where,

D = Day, TL= Online Session

#### 4. Session Coverage

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                     | Deliverables' Mapping                                                                                                   |
|-------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1           | EHACK- TL1    | <p>All the topics as listed below from Session 1 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 1 – Introduction to Ethical Hacking</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and describe Ethical Hacking</li> <li>➤ Describe Attack Modelling</li> <li>➤ List and explain different methodologies of Ethical Hacking</li> </ul>                                  | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 1<br/>XP - Session 1<br/>TG - Session 1</p> |
| 2           | EHACK- TL2    | <p>All the topics as listed below from Session 2 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 2 – Networking Foundations</u></b></p> <ul style="list-style-type: none"> <li>➤ Describe different communication models and Topologies</li> <li>➤ List and describe physical Networks and other Protocols</li> <li>➤ Explain Network Architectures and Cloud Computing</li> </ul>    | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 2<br/>XP - Session 2<br/>TG - Session 2</p> |
| 3           | EHACK- TL3    | <p>All the topics as listed below from Session 3 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 3 – Security Foundations</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain Triad, Information assurance and risk</li> <li>➤ Define and describe Policies, Standards and Procedures</li> <li>➤ Explain different Organising Protections and Security Technology</li> </ul> | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 3<br/>XP - Session 3<br/>TG - Session 3</p> |
| 4           | EHACK – TL4   | <p>The <b>Try It Yourself</b> questions from Session 1 to Session 3 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                                                    | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>Session 1-3</p>                                          |
| 5           | EHACK- TL5    | <p>All the topics as listed below from Session 4 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 4 – Foot Printing and Reconnaissance</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain OSI and DNS</li> </ul>                                                                                                                                                             | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 4<br/>XP - Session 4<br/>TG - Session 4</p> |

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                            | Deliverables' Mapping                                                                                                   |
|-------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Define and describe Passive Reconnaissance, Website Intelligence and Technology Intelligence</li> </ul>                                                                                                                                                                                                                                                                           |                                                                                                                         |
| 6           | EHACK – TL6   | <p>All the topics as listed below from Session 5 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 5 – Scanning Networks</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and explain Ping Sweeps, Port and Vulnerability scanning</li> <li>➤ Describe Packet Crafting and Manipulation</li> <li>➤ Explain different evasion techniques</li> </ul>  | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 5<br/>XP - Session 5<br/>TG – Session 5</p> |
| 7           | EHACK – TL7   | <p>All the topics as listed below from Session 6 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 6 – Enumeration</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and describe service enumeration, remote procedure calls and server message block</li> <li>➤ Explain SNMP and SMTP protocols</li> <li>➤ Define Web-based enumeration</li> </ul> | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 6<br/>XP - Session 6<br/>TG – Session 6</p> |
| 8           | EHACK – TL8   | <p>The <b>Try It Yourself</b> questions from Session 4 to Session 6 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                           | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>Session 4-6</p>                                          |
| 9           | EHACK – TL9   | <p>All the topics as listed below from Session 7 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 7 – System Hacking</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain systems hacking techniques</li> <li>➤ Define and describe client-side vulnerabilities, fuzzing and post exploitation</li> </ul>                                             | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 7<br/>XP - Session 7<br/>TG – Session 7</p> |
| 10          | EHACK – TL10  | <p>All the topics as listed below from Session 8 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 8 – Malware Automated Deployment</u></b></p> <ul style="list-style-type: none"> <li>➤ Define different Malware types and analysis</li> <li>➤ Describe creation of Malware and Malware infrastructure</li> </ul>                                             | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 8<br/>XP - Session 8<br/>TG – Session 8</p> |

| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                   | Deliverables' Mapping                                                                                                      |
|-------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Explain different antivirus solutions</li> </ul>                                                                                                                                                                                                                                                                                                                         |                                                                                                                            |
| 11          | EHACK – TL11  | <p>All the topics as listed below from Session 9 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 9 – Sniffing</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and explain Packet capture and detecting sniffers</li> <li>➤ Explain packet analysis and spoofing attacks</li> </ul>                                                      | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 9<br/>XP - Session 9<br/>TG - Session 9</p>    |
| 12          | EHACK – TL12  | <p>The <b>Try It Yourself</b> questions from Session 7 to Session 9 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                  | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>Session 7-9</p>                                             |
| 13          | EHACK – TL13  | <p>All the topics as listed below from Session 10 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 10 – Social Engineering</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain Social Engineering</li> <li>➤ Define and describe the Phishing and Website attacks</li> <li>➤ Describe Wireless and Automating Social Engineering</li> </ul> | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 10<br/>XP - Session 10<br/>TG - Session 10</p> |
| 14          | EHACK – TL14  | <p>All the topics as listed below from Session 11 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 11 – Wireless Security</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and describe different wireless security</li> <li>➤ Explain Wi-Fi, bluetooth and mobile devices security</li> </ul>                                            | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 11<br/>XP - Session 11<br/>TG - Session 11</p> |
| 15          | EHACK – TL15  | <p>All the topics as listed below from Session 12 of A <i>Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 12 – Attack and Defense</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain Web application attacks and DoS attacks</li> </ul>                                                                                                           | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 12<br/>XP - Session 12<br/>TG - Session 12</p> |



| Session No. | Session Title | Session Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Deliverables' Mapping                                                                                                      |
|-------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|             |               | <ul style="list-style-type: none"> <li>➤ Define and describe application exploitation and lateral movement</li> <li>➤ Define Defense in Depth/Defense in Breath and Defensible Network Architecture</li> </ul>                                                                                                                                                                                                                                                                                                                             |                                                                                                                            |
| 16          | EHACK – TL16  | <p>The <b>Try It Yourself</b> questions from Session 10 to Session 12 of <i>A Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p>                                                                                                                                                                                                                                                                                                                                                                         | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>Session 10-12</p>                                           |
| 17          | EHACK – TL17  | <p>All the topics as listed below from Session 13 of <i>A Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 13 – Cryptography</u></b></p> <ul style="list-style-type: none"> <li>➤ Define Encryption and Cryptography</li> <li>➤ Define and describe Certificate Authorities and Key Management</li> <li>➤ Explain Cryptographic Hashing, Pretty Good Privacy (PGP) and Secured Multipurpose Internet Mail Extensions (S/MIME)</li> <li>➤ Describe Disk and File Encryption</li> </ul> | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 13<br/>XP - Session 13<br/>TG – Session 13</p> |
| 18          | EHACK – TL18  | <p>All the topics as listed below from Session 14 of <i>A Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 14 – Security Architecture and Design</u></b></p> <ul style="list-style-type: none"> <li>➤ Explain different data classification and security models</li> <li>➤ Define and describe application and security architecture</li> </ul>                                                                                                                                       | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 14<br/>XP - Session 14<br/>TG – Session 14</p> |
| 19          | EHACK – TL19  | <p>All the topics as listed below from Session 15 of <i>A Complete Guide to Ethical Hactivists</i> book should be covered in this session.</p> <p><b><u>Session 15 – Cloud Computing and the Internet of Things (IoT)</u></b></p> <ul style="list-style-type: none"> <li>➤ Define and describe cloud computing architecture and deployment</li> <li>➤ Explain common cloud threats</li> <li>➤ Define and explain the IOT and OT</li> </ul>                                                                                                 | <p><b><u>A Complete Guide to Ethical Hactivists</u></b></p> <p>SG - Session 15<br/>XP - Session 15<br/>TG – Session 15</p> |

| Session No. | Session Title | Session Details                                                                                                                                             | Deliverables' Mapping                                                     |
|-------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 20          | EHACK – TL20  | The <b>Try It Yourself</b> questions from Session 13 to Session 15 of <i>A Complete Guide to Ethical Hactivists</i> book should be covered in this session. | <b><u>A Complete Guide to Ethical Hactivists</u></b><br><br>Session 13-15 |

## 5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

### ➤ Deliverables for Students:

|                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| <b>Learner's Guide</b>    | eBook consisting of the knowledge to be imparted (wherever applicable)  |
| <b>Assignments</b>        | Case study/scenario based problems to be solved by learners             |
| <b>Article</b>            | Additional knowledge in the form of articles that are referred from Web |
| <b>Practice Questions</b> | Additional practice questions for learners at the end of each session   |

| Module          | Book Code          | Book Name                                  |
|-----------------|--------------------|--------------------------------------------|
| KIT             | OV-1525-KIT01-01-A | Kit of 1 title for OV-1525 Ethical Hacking |
| Ethical Hacking | AINNEHACK10224E000 | A Complete Guide for Ethical Hactivists    |

### ➤ Deliverables for Faculty

|                                |                                                                            |
|--------------------------------|----------------------------------------------------------------------------|
| <b>Presenter's Manual</b>      | A training delivery plan for the trainers                                  |
| <b>Trainer's Guide</b>         | A session wise guide for the trainers                                      |
| <b>Classroom Presentations</b> | Session wise PowerPoint presentations as teaching aid, wherever applicable |

| Module Name     | Book Code          | Book Name                                  |
|-----------------|--------------------|--------------------------------------------|
| KIT             | OV-1525-KIT01-01-A | Kit of 1 title for OV-1525 Ethical Hacking |
| Ethical Hacking | AINNEHACK10224E000 | A Complete Guide for Ethical Hactivists    |

## 6. Library References

- **CEH v12 Certified Ethical Hacker Study Guide** by Ric Messier
- **The Ethical Hacking Book for Beginners: A Step by Step Guide for you to Learn the Fundamentals of Ethical Hacking and CyberSecurity (CyberSecurity and Hacking)** by Ramon Nastase
- **The Complete Ethical Hacking Book: A Comprehensive Beginner's Guide to Learn and Master in Ethical Hacking** by Thirumalesh

## 7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination (Objective)
- Practical Examination

### 7.1 Award of Grades

**Overall Weighted Marks > = 40% but < 60%** qualifies for **PASS**

**Overall Weighted Marks > = 60% but < 75%** qualifies for **CREDIT** and

**Overall Weighted Marks > = 75%** qualifies for **DISTINCTION**

**Note:** To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of the Examination; otherwise he/she will be declared as 'Referred'.

### 7.2 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

#### Final Examination 1 (E1)

**Marks:** 50

**Duration:** 1 hour

**Type:** Objective

| S# | Name of the Subject | Marks |
|----|---------------------|-------|
| 1. | Ethical Hacking     | 50    |

#### Final Examination 2 (R1)

**Marks:** 50

**Duration:** 2 hours

**Type:** Practical

| S# | Name of the Subject | Marks |
|----|---------------------|-------|
| 1. | Ethical Hacking     | 50    |

### Evaluation Details

**OES-50%**

**Practical-50%**

## **8. Classroom and Lab Pre-requisites**

---

The requirements for conducting the sessions have been described as follows:

### **8.1 Software Requirements**

Microsoft Windows 11  
Ethical Hacking Tools

### **8.2 Recommended Hardware**

Intel Core i5/i7 Processor or higher  
8 GB or 16 GB RAM  
1 TB Hard Disk space  
Mouse  
Keyboard

### **8.3 Other Requirements (For Classroom)**

Computer for giving demonstrations wherever applicable  
LCD projector

~~ End of Document ~~